

Physics I

Code : EBA1110

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LECTURE 7

Capacitors

Capacitor definition

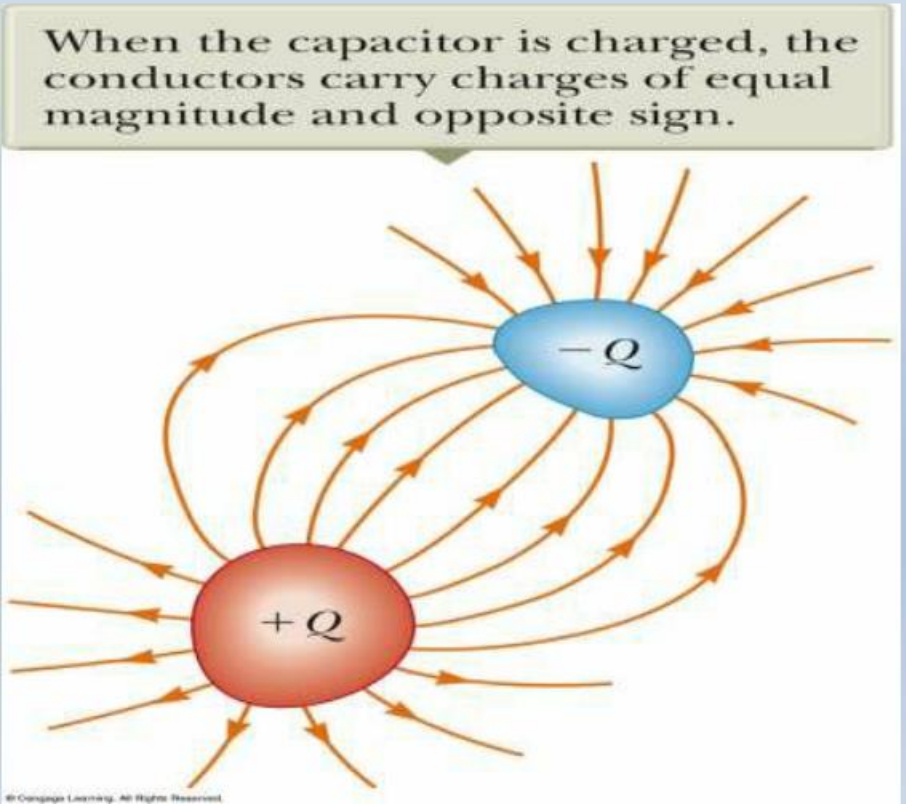
Capacitors are devices that store electric charge.

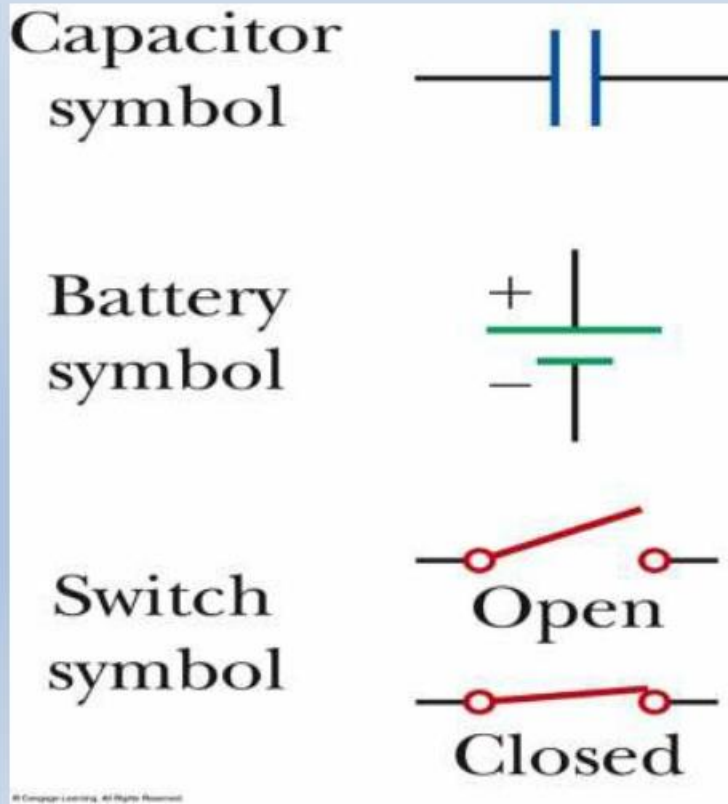
Examples of where capacitors are used include:

- radio receivers
- filters in power supplies
- energy-storing devices in electronic flashes



- A capacitor consists of two conductors.
- These conductors are called plates.
- When the conductor is charged, the plates carry charges of equal magnitude and opposite directions.
- A potential difference exists between the plates due to the charge.





Definition of Capacitance

The **capacitance**, *C*, of a capacitor is defined as the ratio of the magnitude of the charge on either conductor to the potential difference between the conductors.

$$C \equiv \frac{Q}{\Delta V}$$

- The capacitance is a measure of the capacitor's ability to store charge .
- The capacitance of a capacitor is the amount of charge the capacitor can store per unit of potential difference.

- The SI unit of capacitance is the **farad** (F).
- The farad is a large unit, typically you will see microfarads (μF) and picofarads (pF).

Capacitance

$$C \equiv \frac{Q}{\Delta V}$$

Parallel Plates

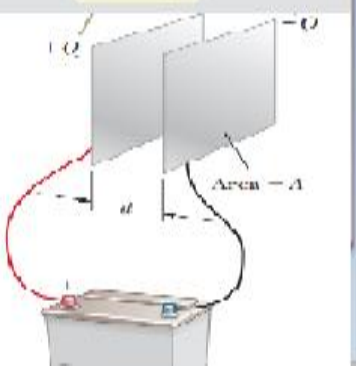
The charge density on the plates is

$\sigma = Q/A$.

The capacitance is proportional to the area of its plates and inversely proportional to the distance between the plates.

$$C = \frac{Q}{\Delta V} = \frac{Q}{Qd/\epsilon_0 A}$$

$$C = \frac{\epsilon_0 A}{d}$$

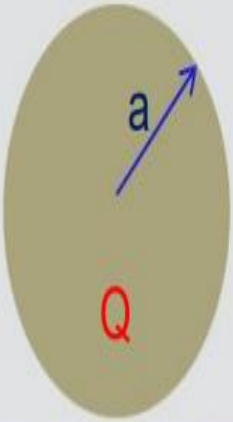


Isolated Sphere

Assume a spherical charged conductor with radius *a*.

$$C \equiv \frac{Q}{\Delta V}$$

$$= \frac{Q}{k_e Q/a} = \frac{R}{k_e} = 4\pi\varepsilon_0 a$$



a Cylindrical Capacitor

The capacitance is

$$C \equiv \frac{Q}{\Delta V}$$

$$= \frac{\ell}{2k_e \ln(b/a)}$$



a Spherical Capacitor

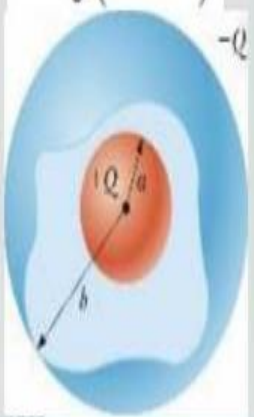
The potential difference will be

$$\Delta V = k_e Q \left(\frac{1}{b} - \frac{1}{a} \right)$$

The capacitance will be

$$C \equiv \frac{Q}{\Delta V}$$

$$= \frac{ab}{k_e (b-a)}$$



Example 1

A parallel-plate capacitor with air between the plates has an area $A = 2.00X10^{-4}\ m^2$ and a plate separation d=1.00 mm. Find its capacitance.

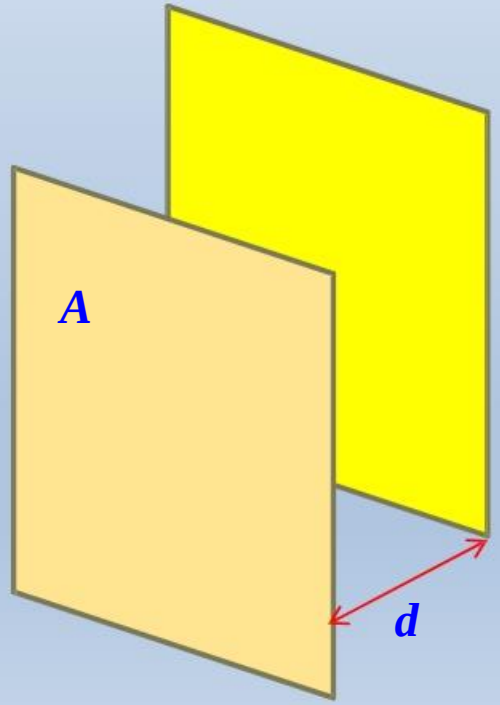
Solution:

$$C = \frac{\varepsilon_o A}{d}$$

$$C = \frac{(8.85 \times 10^{-12})(2 \times 10^{-4})}{1 \times 10^{-3}}$$

$$C = 1.77 \times 10^{-12} F$$

$$C = 1.77\ pF$$



Energy Stored in a Capacitor

Assume the capacitor is being charged and, at some point, has a charge q on it.

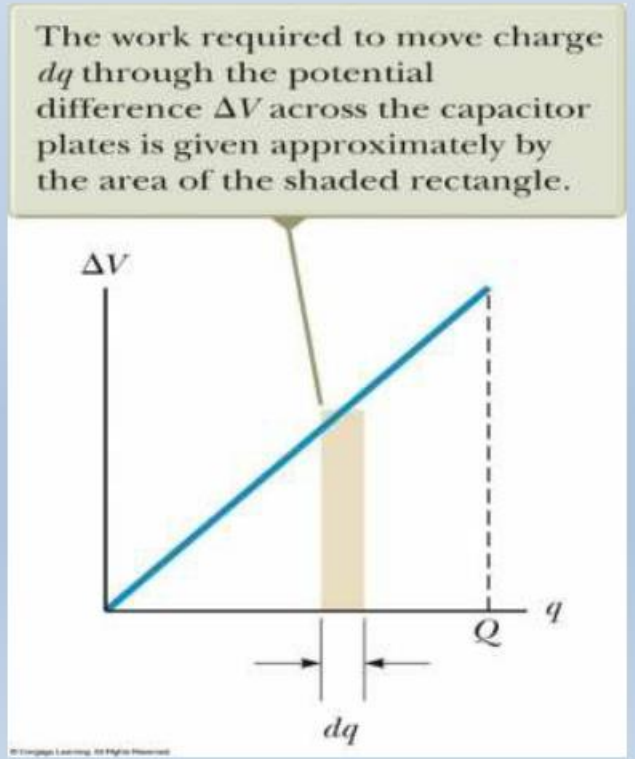
The work needed to transfer a charge from one plate to the other is

$$dW = \Delta V dq = \frac{q}{C} dq$$

The work required is the area of the tan rectangle.

The total work required is

$$W = \int_0^Q \frac{q}{C} dq = \frac{Q^2}{2C}$$



The work done in charging the capacitor appears as electric potential energy *U*

$$U = \frac{Q^2}{2C} = \frac{1}{2} Q \Delta V = \frac{1}{2} C (\Delta V)^2$$

- This applies to a capacitor of any geometry.
- The energy stored increases as the charge increases and as the potential difference increases.
- In practice, there is a maximum voltage before discharge occurs between the plates.
- The energy can be considered to be stored in the electric field .
- For a parallel-plate capacitor, the energy can be expressed in terms of the field as $U = \frac{1}{2} (\varepsilon_o A d) E^2$
- It can also be expressed in terms of the energy density (energy per unit volume) $U_E = \frac{1}{2} \varepsilon_o E^2$.

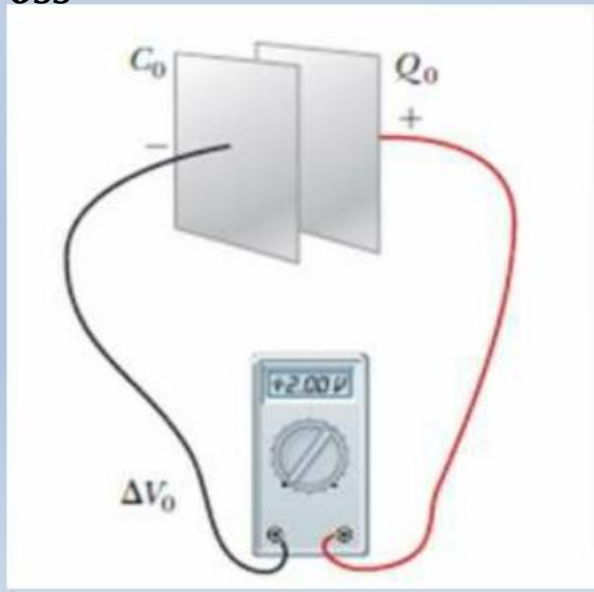
Capacitors with Dielectrics

A dielectric is a nonconducting material that, when placed between the plates of a capacitor, increases the capacitance.

such as rubber, glass, and waxed paper

Consider parallel-plate capacitor that without a dielectric has a charge Q_0 and a capacitance C_0. The potential difference across the capacitor is

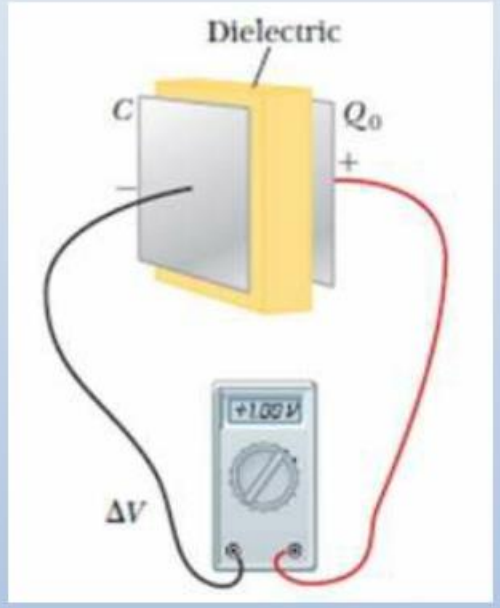
$$V_o = \frac{Q_o}{C_o}$$



If a dielectric is now inserted between the plates as in Figure, the voltmeter indicates that the voltage between the plates decreases to a value V. The voltages with and without the dielectric are related by a factor k as follows:

$$V = \frac{V_o}{K}$$

K is dielectric constant



$$C = \frac{Q_o}{V} = \frac{Q_o}{V_o/K} = K\frac{Q_o}{V_o}$$

With a dielectric, the capacitance becomes

$$C = KC_o$$

$$C = K\varepsilon_o\frac{A}{d}$$

The capacitance increases by the factor κ when the dielectric completely fills the region between the plates.

Some Dielectric Constants and Dielectric Strengths :

Table 26.1 Approximate Dielectric Constants and Dielectric Strengths of Various Materials at Room Temperature

Material	Dielectric Constant κ	Dielectric Strength[a] (10^6 V/m)
Air (dry)	1.000 59	3
Bakelite	4.9	24
Fused quartz	3.78	8
Mylar	3.2	7
Neoprene rubber	6.7	12
Nylon	3.4	14
Paper	3.7	16
Paraffin-impregnated paper	3.5	11
Polystyrene	2.56	24
Polyvinyl chloride	3.4	40
Porcelain	6	12
Pyrex glass	5.6	14
Silicone oil	2.5	15
Strontium titanate	233	8
Teflon	2.1	60
Vacuum	1.000 00	—
Water	80	—

[a]The dielectric strength equals the maximum electric field that can exist in a dielectric without electrical breakdown. These values depend strongly on the presence of impurities and flaws in the materials.

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Example 2

A capacitor has plates of area 0.034m2 and a separation of 2mm in air. The potential difference between the plates is 200 V.

a) Calculate the capacitance.

b) Calculate the electric field between the plates.

c) Calculate the charge on each plate.

Solution

a) Calculate the capacitance.

$$C = \frac{\varepsilon_o A}{d}$$

$$C = \frac{(8.85 \times 10^{-12})(0.034)}{2 \times 10^{-3}} = 150.45 \times 10^{-12} F = 150.45 pF$$

b) Calculate the electric field between the plates.

$$E = \frac{V}{d} = \frac{200}{2 \times 10^{-3}} = 10^5 volt/m$$

c) Calculate the charge on each plate.

$$Q = CV$$

$$Q = (150.45 \times 10^{-12})(200)$$

$$Q = 30.1 \times 10^{-9} C$$

Example 3

A parallel plate capacitor has an area of 2000 cm^2 and separation of 1 cm . The original potential difference between the plates, V_o, is 3000 V , and it decreases to 1000 V when a sheet of dielectric is inserted between the plates. Compute:

a) The original capacitance C_o.
b) The charge Q on the plates.
c) The capacitance C after insertion of the dielectric.
d) The dielectric constant κ of the dielectric.
e) The permittivity ε of the dielectric.
f) The original electric field E_0 between the plates.
g)The electric field E after insertion of the dielectric.

Solution

A=2000 cm^2 *d=1 cm* *V_o =3000 volt* *V=1000 volt*

a) The original capacitance C .

$$C_o = \varepsilon_o \frac{A}{d} = 8.85 \times 10^{-12} \frac{0.2}{0.01} = 1.77 \times 10^{-10} F$$

b) The chargeQ on the plates.

$$Q = C_o V_o = 1.77 \times 10^{-12} \times 3000 = 531 \times 10^{-9} C$$

c) The capacitance C after insertion of the dielectric.

$$C = \frac{Q}{V} = \frac{531 \times 10^{-9}}{1000} = 531 \times 10^{-12} F$$

d) The dielectric constant κ of the dielectric.

$$K = \frac{C}{C_o} = \frac{531 \times 10^{-12}}{177 \times 10^{-12}} = 3$$

e) The permittivity ε of the dielectric.

$$\varepsilon = k\varepsilon_o = 3 \times 8.85 \times 10^{-12} = 26.6 \times 10^{-12} \frac{C}{m^2}.N$$

f) The original electric field E_0 between the plates.

$$E_o = \frac{\sigma}{\varepsilon_o} = \frac{Q}{A\varepsilon_o} = \frac{531 \times 10^{-9}}{8.85 \times 10^{-12}} = 3 \times 10^5 \ V/m$$

g) The electric field E after insertion of the dielectric

$$E = \frac{E_o}{K} = \frac{3 \times 10^5}{3} = 10^5 \ V/m$$